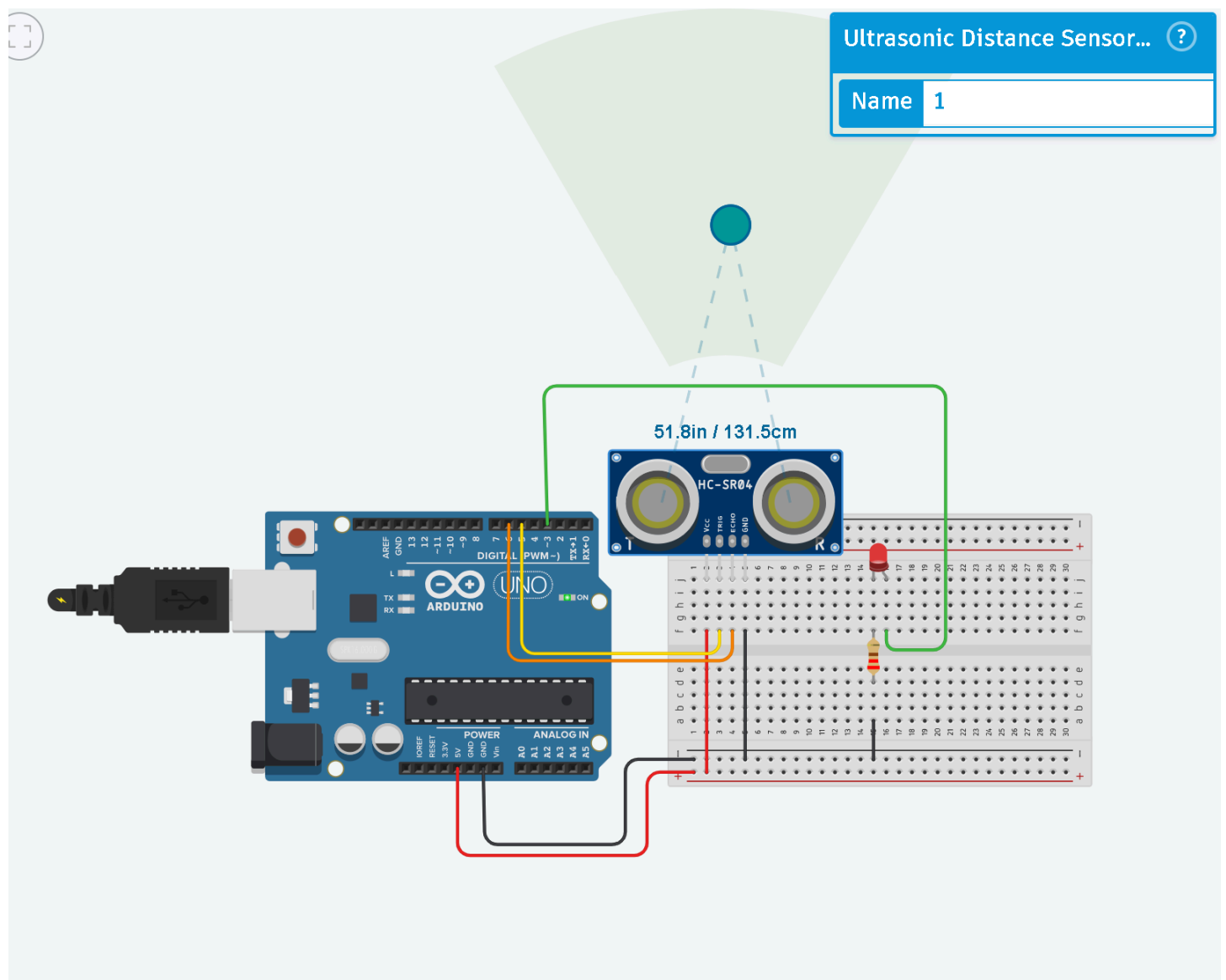




Source Code

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Assignment_1_day_6.cpp

Cpp_programs\Robocon_programs\Assignment_1_day_6.cpp

```
1  /*
2  Code by Rayan Raphy
3  */
4
5  //declaring the pins
6  const int trig = 5;
7  const int echo = 6;
8
9  const int led = 3;
10
11 //custom map funciton:
12 float mapper(double value,double ini_min,double ini_max,double fin_min,double fin_max){
13     if(value<ini_min || value > ini_max){
14         return 0; // outside limit cases handled in this seciton
15     }else{
16         return (value - ini_min)*(fin_max - fin_min)/(ini_max - ini_min); //mapping is done here
17     }
18 }
19 void setup()
20 {
21     //declaring the input and output pins
22     pinMode(led, OUTPUT);
23
24     pinMode(trig, OUTPUT);
25     pinMode(echo, INPUT);
26
27     Serial.begin(9600);
28 }
29
30 void loop()
31 {
32     //acquiring duration from hiding ht eobstacle
33     digitalWrite(trig,LOW);
34     delayMicroseconds(2);
35     //clearing the trig pin of outputs
36
37     digitalWrite(trig,HIGH);
38     delayMicroseconds(10);
39     digitalWrite(trig,LOW);
40     //fired the sound wave
41     long duration = pulseIn(echo,HIGH);
42
43 }
```

```
43 |
44 | //waiting untill the signal reach the echo
45 | float distance = (duration*0.0343)/2.0;
46 | int bright = mapper(distance,20,200,0,255);
47 | //calculating the distance and mapping the
48 |
```

localhost:26793/78528d9b-1760-4773-ba61-897411121185/

1/2

9/16/26, 10:39 PM

Assignment_1_day_6.cpp

```
49 | analogWrite(led,bright);
50 |
51 | Serial.print("Distance: ");
52 | Serial.print(distance);
53 | Serial.print(" | LED brightness: ");
54 | Serial.println(bright);
55 | }
```

CUSTOM MAP FUNCTION!!

```
11 | //custom map funciton:
12 | float mapper(double value,double ini_min,double ini_max,double fin_min,double fin_max){
13 |     if(value<ini_min || value > ini_max){
14 |         return 0; // outside limit cases handled in this seciton
15 |     }else{
16 |         return (value - ini_min)*(fin_max - fin_min)/(ini_max - ini_min); //maping is done here
17 |     }
18 | }
```

OUTPUT SCREENSHOT!!

Ultrasonic Distance Sensor...

Name 1

51.8In / 131.5cm

HC-SR04

7. Submission:

Submit the following in a single PDF submission:

A. Circuit Screenshot:

- Clear screenshot of the completed circuit.
- Breadboard and all connections must be clear.
- Wiring should be neat and color-coded.

B. Source Code:

- Complete Arduino source code.
- Include the self-written custom map function.
- The custom map function must return FLOAT.
- Do not submit code containing the Arduino IDE comments.

C. The circuit screenshot and source code should be submitted together.

Expected Output:

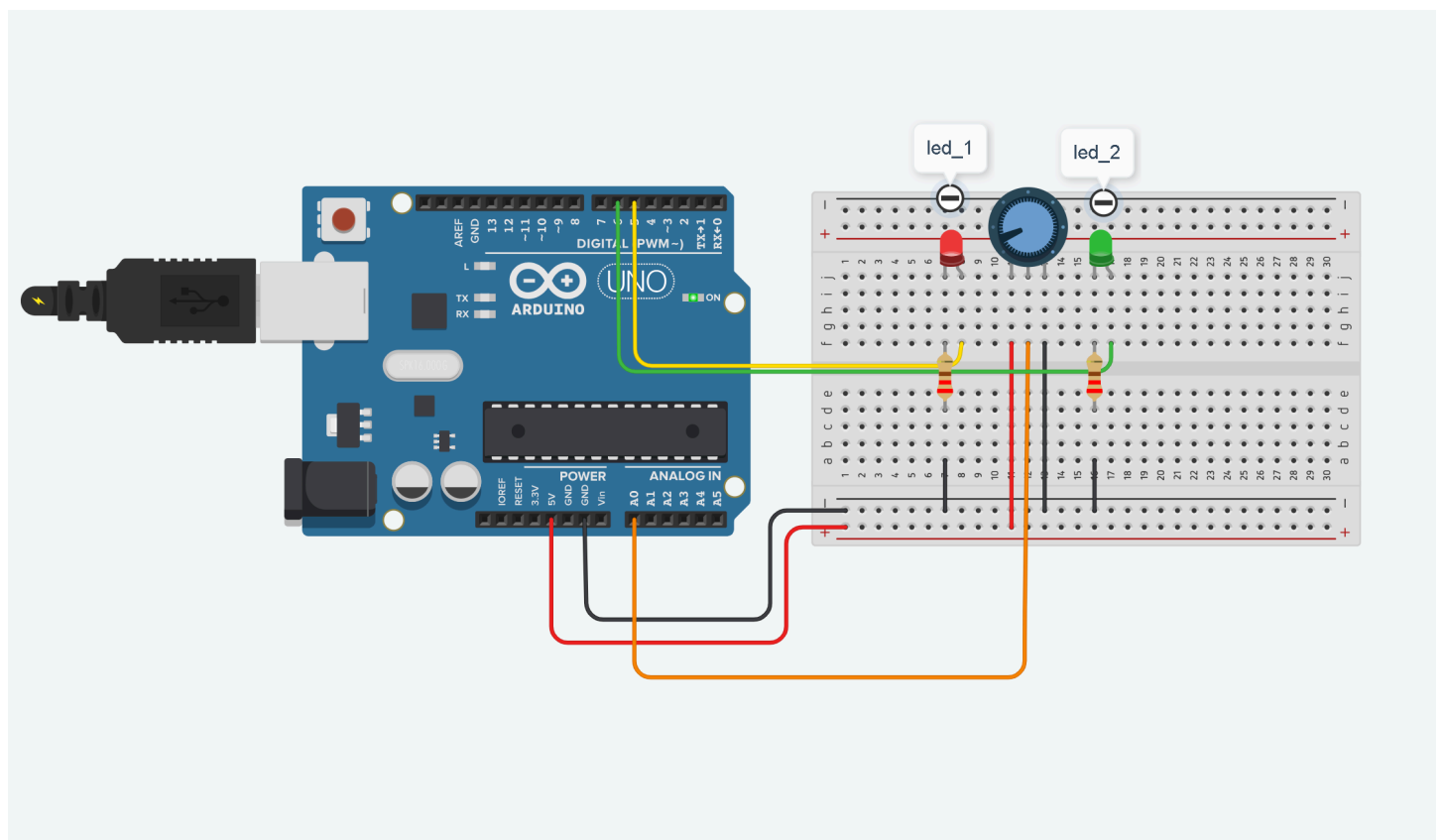
As the object moves farther from the ultrasonic sensor, the LED brightness increases.

Serial Monitor

Distance	LED brightness
Distance: 129.00	LED brightness: 154
Distance: 129.02	LED brightness: 154
Distance: 129.05	LED brightness: 154
Distance: 129.04	LED brightness: 154
Distance: 1	

Assignment -2

Circuit Diagram:



Source Code:

Cpp_programs\Robocon_programs\Assignment_2_day_6.cpp

```
1  /*
2
3
4  code By Rayan Raphy
5  */
6  //initializing the pin
7  const int led_1 = 5;
8  const int led_2 = 6;
9
10 const int pot = A0;
11
12 void setup()
13 {
14     pinMode(led_1, OUTPUT);
15     pinMode(led_2, OUTPUT);
16
17     pinMode(pot, INPUT);
18
19     Serial.begin(9600);
20
21 }
22
23 void loop()
24 {
25     int potVal = analogRead(pot);
26
27     int bright = map(potVal,0,1023,0,255); //converting the 10bit analog inputs to 8bit analog
    outputs
28
29     analogWrite(led_1,bright);
30     analogWrite(led_2,(255-bright)); //supplying th complimentary brigthness to led 2
31
32     Serial.print("Led 1 brightness: ");
33     Serial.print(bright);
34     Serial.print(" | Led 2 brightness: ");
35     Serial.println(255-bright);
36
37 }
```

Output screenshot!!

